Lab 8

DSA

March 20, 2026

Exercise

You are given n floating point numbers in the range $[0, 1)$.
Your task is to implement the Bucket Sort algorithm in C using the provided pseudocode as reference.

Your program should do the following:

1. Create n buckets.
2. Compute the bucket index for each element using $[n \times A[i]]$ and insert the element into the corresponding bucket.
3. Sort each bucket using **Insertion Sort**.
4. Concatenate all buckets to produce the final sorted array.

Example Input

0.78 0.17 0.39 0.26 0.72 0.94 0.21 0.12 0.23 0.68

Expected Output

0.12 0.17 0.21 0.23 0.26 0.39 0.68 0.72 0.78 0.94

Bucket Sort Pseudocode

```
BUCKET-SORT(A, n)
let B[0 ... n − 1] be a new array
for i = 0 to n − 1
    make B[i] an empty linked list
for i = 1 to n
    insert A[i] into list B[⌊n · A[i]⌋]
for i = 0 to n − 1
    sort list B[i] with insertion sort
concatenate lists B[0], B[1], ..., B[n − 1] together in order
return the concatenated list
```

Rubric (10 Marks)

- Correct bucket creation (2 marks)
- Correct bucket index computation (2 marks)
- Correct insertion into buckets (2 marks)
- Sorting elements within buckets (2 marks)
- Correct concatenation and final output (2 marks)